NLP

BERT vs GPT Visual Guide

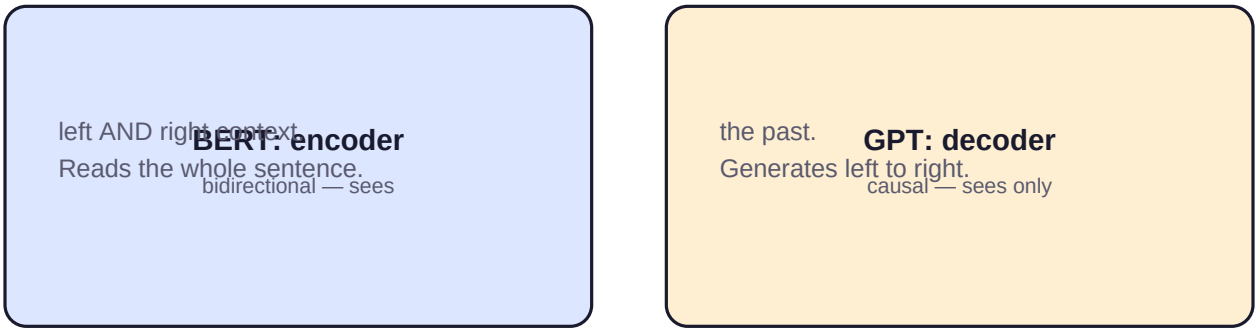
Two pretraining philosophies:
masked language modeling vs next-token prediction.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Encoder vs Decoder

Same transformer blocks, opposite information flow.

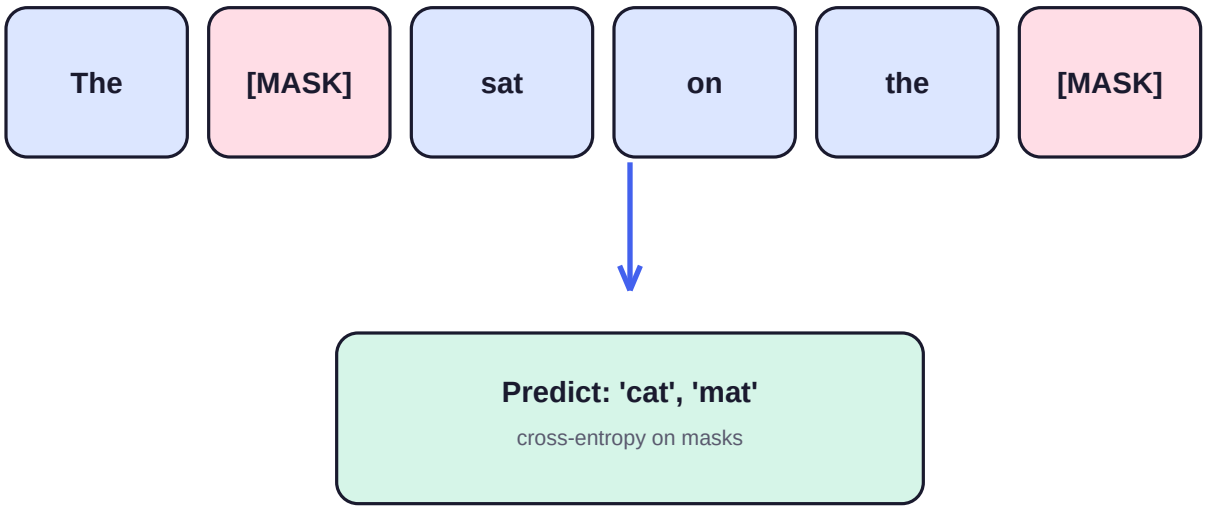


Encoders understand; decoders generate. The masking pattern is the entire difference.

STEP 2/5

Masked LM: BERT's Pretraining

Hide words, predict them from both sides.



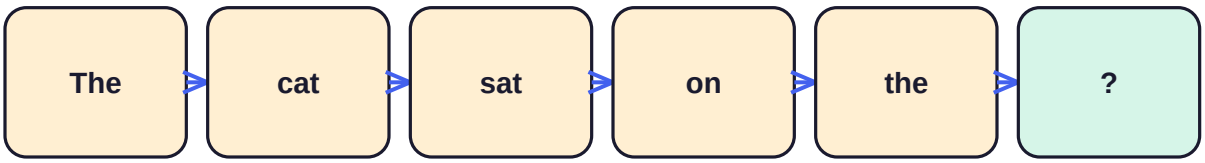
Mask ~15% of tokens at random. Loss is computed ONLY on the masked positions.

Bidirectional context makes BERT superb at understanding — classification, NER, QA span extraction.

STEP 3/5

Next-Token: GPT's Pretraining

Predict the next word, forever.



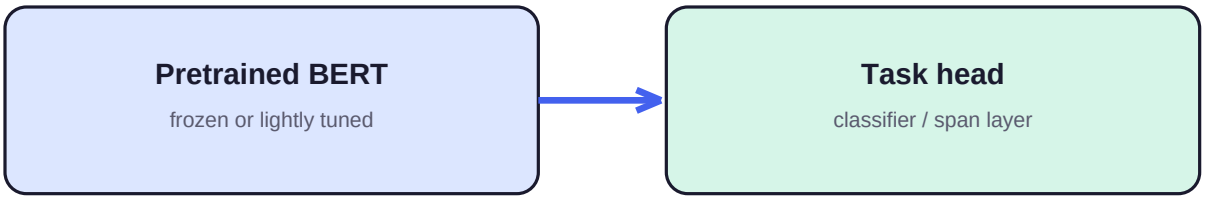
Every position predicts the next token. The entire internet becomes a self-supervised dataset.

Simple objective, staggering scale: next-token prediction at trillion-token scale yields general capabilities.

STEP 4/5

Fine-tuning BERT

Add a small head, train on labeled data.



- Classification: feed the [CLS] embedding into a linear layer (sentiment, intent, topic).
- QA: two heads predict answer start/end spans over the passage.
- Fine-tune all layers with a small learning rate (2e-5) for a few epochs — watch for overfitting on small sets.

STEP 5/5

Adapting GPT

Prompt it, or fine-tune it.

Prompting

zero/few-shot + instructions

Fine-tuning

train on your Q/A pairs

- Prompting first: cheapest adaptation. Good prompts often beat small fine-tunes.
- Fine-tune when you need a stable style, domain vocabulary, or structured outputs at scale.
- RLHF/DPO aligns the model with human preferences after pretraining — the chat behavior you know.
- Rule of thumb: prompt for behavior, fine-tune for knowledge and format reliability.

STEP ✓/✓

Key Takeaways

BERT vs GPT — Remember

- BERT = bidirectional encoder trained with masked LM: best for understanding tasks.
- GPT = causal decoder trained with next-token prediction: best for generation.
- Masking pattern is the core difference; architecture is otherwise similar.
- Fine-tune BERT with a task head; adapt GPT via prompting first, fine-tuning second.
- Both benefit from domain-adaptive pretraining before task fine-tuning.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev